MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED, M7
METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated, M7 Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MIL-A-2550	- Ammunition and Special Weapons, General Specification For

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings

DRAWINGS

Armament Command

F8593278	- Thruster, Cartridge Actuated, M7, Metal Parts Assembly
C8796480	- Box, Fiberboard (Inner), Packing Ammunition For Thruster, Cartridge Actuated, M7, with Cartridge Impulse M67A1
C8796479	- Box, Packing, Ammunition for Thruster, Cartridge Actuated, M7, with Cartridge Impulse M67A1

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2.1 (Cont'd)

DRAWINGS (Cont'd)

IEL-8593278	- Inspection Equipment List for Thruster, M7, Metal Parts Assembly
C9234882	- Box, Fiberboard (Outer) Alternative Packing Ammunition for Thruster, C/A, M7 with Cartridge Impulse M67A1

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with all the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Mechanical properties.-Mechanical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.3 Hydrostatic pressure.-The body and head shall withstanda a constant internal pressure of 5,000 lbs per square inch (psi), minimum, for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure. (See 6.1).

3.6 Structural strength.- The assembled thruster shall withstand a static tensile force of 500 lbs, minimum, for 15 seconds, minimum, between mounting points without unlocking and without mechanical failure. (See 6.1)

3.7 Metal parts assembly.-Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing.

3.7.1 All threaded joints shall be assembled handtight.

3.7.2 End sleeve rotation.-Upon completion of assembly and after the structural strength test, the end sleeve shall be free to rotate 360 degrees in a clockwise direction.

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3.8 Protective finish.-Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.9 Residual magnetism.-The metal parts assembly shall not deflect the indicator of an approved compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.10 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.10.1 Metal defects.-All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.10.2 Burr.-No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.3 Foreign matter.-No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications and drawing where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

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4.2 First Article Sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 10 sets of unassembled components for dimensional and visual examination and tests, and 50 shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provision acceptance at source, the sample shall be inspected for all requirements of the drawings, and specifications at a Government laboratory or such other facilities specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Product" provisions of MIL-STD-105.

4.3.1.2 Size of lot.- The shipping lot of metal parts assemblies shall consist of 1200 Thrusters, maximum, provided the homogeneity requirements of inspection lots have been retained. A small lot may be shipped upon approval of the contracting officer.

4.3.2 Examination.-One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and associated criteria in Tables I and II. For an individual defect (single characteristic) the criteria in Table I shall be used. For total defects within a major or minor group the criteria in Table II shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed in ANSI B46.1 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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4.3.2 (Cont'd)

TABLE I

Lot Size	Sample Size	Major Accept	Major Reject	Minor Accept	Minor Reject
1 - 300	100 Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE II

	Total allowable defects per sample size			
Sample Size for Major Defects	110	150	225	300
Number of Characteristics in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Size for Minor Defects	**110**	**150**	**225**	**300**
Number of Characteristics in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
6 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Head, Drawing F8593279, covering a detail of Drawing F8593278.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of large thread, max.	Gage
102. Minor diamter of large thread, max.	Gage

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4.3.2.1 (Cont'd)

	Defect	Method of Inspection
MAJOR (Cont'd)		
103.	Pitch diameter of thread, small end, max.	Gage
104.	Minor diameter of thread, small end, max.	Gage
105.	Concentricity of cavity counterbore with large thread	Gage
106.	Concentricity of cavity with large thread	Gage
107.	True position of lugs	Gage
108.	Concentricity of lugs	Gage
109.	Symmetry of flats at root of lugs with cavity	Gage
110.	Diameter of cavity, max.	Gage
111.	Diameter of cavity, counterbore, max.	Gage
112.	Diameter of lugs, min.	Gage
113.	Diameter of lug flange	Gage
114.	Depth of cavity	Gage
115.	Depth of cavity counterbore	Gage
116.	Depth of counterbore, small end	Gage
117.	Length of effective thread, large end, min.	Gage
118.	Length of effective thread, small end, min.	Gage
119.	Length of ineffective thread, large end, max.	Gage
120.	Length of ineffective thread, small end, max.	Gage
121.	Width across flats of lugs, min.	Gage
122.	Location of lugs	Gage
123.	Radius at base of lugs	Measure
124.	Diameter of chamfer, small end	Measure
125.	Angle of chamfer at small end	Visual
126.	Thread damaged or not full	Visual
127.	Metal defective	Visual
128.	Protective finish inadequate or defective	Visual
129.	Surface finish of small thread face and chamfer, improper	Visual
130.	Burr or sharp corner	Visual
131.	Foreign matter	Visual
MINOR		
201.	Pitch dimater of large thread, min.	Gage
202.	Pitch diameter of thread, small end, min.	Gage
203.	Pitch diameter of set screw thread	Gage
204.	Minor diameter of set screw thread, max.	Gage
205.	Diameter of cavity, min.	Gage
206.	Diameter of lugs, max.	Gage
207.	Diameter of cavity counterbore, min.	Gage
208.	Diameter of counterbore, small end	Gage

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
209. Diameter of body	Gage
210. Diameter of flange at set screw holes	Gage
211. Diameter of hole through midsection	Gage
212. Length to lug flange	Gage
213. Width of lug flange	Gage
214. Width across flats of lugs, max.	Gage
215. Diameter of shoulder at small end	Measure
216. Depth of flats at small end	Measure
217. Depth of cavity, small end	Measure
218. Length to shoulder at small end	Measure
219. Length of lugs	Measure
220. Length of flat at small end	Measure
221. Total length	Measure
222. Width of flats at lugs	Measure
223. Width of flange at set screw holes	Measure
224. Location of set screw holes	Measure
225. Fillets or corners incorrect	Visual
226. Marking missing, incorrect or illegible	Visual
227. Radius or chamfer, missing or incorrect	Visual
228. Surface finish, improper, except as otherwise classified	Visual

4.3.2.2 Body, Drawing D8593280, covering a detail of Drawing F8593278.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of mouth with pitch diameter of thread	Gage
104. Concentricity of thread pilot with pitch diameter of thread	Gage
105. Concentricity of counterbore with pitch diameter of thread	Gage
106. Concentricity of long cavity with pitch diameter of thread	Gage
107. Concentricyt of short cavity with long cavity	Gage

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4.3.2.2 (Cont'd)

Defect		Method of Inspection
MAJOR (Cont'd)		
108.	Concentricity of throat with long cavity	Gage
109.	Concentricity of groove with short cavity	Gage
110.	Concentricity of counterbore, with short cavity	Gage
111.	Diameter of long cavity, max.	Gage
112.	Diameter of throat, max.	Gage
113.	Diameter of short cavity	Gage
114.	Diameter of groove	Gage
115.	Diameter of counterbore, groove end, max.	Gage
116.	Diameter of mouth at threaded end	Gage
117.	Diameter of counterbore, threaded end	Gage
118.	Diameter of thread pilot, min.	Gage
119.	Depth from threaded end to throat	Gage
120.	Depth to bottom of counterbore, threaded end	Gage
121.	Thickness of wall, min.	Gage
122.	Location of groove	Gage
123.	Internal radii forwarded end, long cavity	Gage
124.	Straightness of bore	Gage
125.	Thread damaged or not full	Visual
126.	Sharp corner(s) missing at throat	Visual
127.	Metal defective	Visual
128.	Protective finish inadequate or defective	Visual
129.	Surface finish, on long cavity and throat, improper	Visual
130.	Burr or sharp corner, except as otherwise classified	Visual
131.	Foreign matter	Visual
MINOR		
201.	Pitch diamete/of thread, max.	Gage
202.	Diameter of long cavity, min.	Gage
203.	Diameter of throat, min.	Gage
204.	Diameter of short cavity, min.	Gage
205.	Diameter of counterbore, groove end, min.	Gage
206.	Diameter of thread pilot, max.	Gage
207.	Diameter of body at midsection, max.	Gage
208.	Depth of counterbore, groove end, min.	Gage
209.	Total length	Gage
210.	Radius at mouth cavity	Gage
211.	Diameter of body, groove end	Measure
212.	Depth of mouth at thread end	Measure

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
213. Depth of short cavity	Measure
214. Length to end of thread	Measure
215. Length of thread pilot	Measure
216. Length of body, groove end	Measure
217. Profile of groove	Measure
218. Wrench flats, incorrect	Visual
219. Marking missing, incorrect or illegible	Visual
220. Radius or chamfer missing or incorrect, except as otherwise classified	Visual
221. Surface finish, improper, except as otherwise classified	Visual

4.3.2.3 Sleeve, End, Drawing D8593281, covering a detail of Drawing F8593278.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of shoulder with body	Gage
104. Concentricity of cavity with body	Gage
105. Concentricity of pitch diameter of thread with body	Gage
106. Symmetry of slots with body	Gage
107. Diameter of body, slotted end, min.	Gage
108. Diameter of cavity, max.	Gage
109. Diameter of shoulder, min.	Gage
110. Width of slots	Gage
111. Location of slots	Gage
112. Thread damaged or not full	Visual
113. Metal defective	Visual
114. Protective finish inadequate or defective	Visual
115. Surface finish of slots, improper	Visual
116. Burr	Visual
117. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, min.	Gage
202. Diameter of body, slotted end, max.	Gage
203. Diameter of cavity, min.	Gage
204. Diameter of shoulder, max.	Gage
205. Depth of cavity	Gage

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4.3.2.3 (Cont'd)

	Defect	Method of Inspection
MINOR (Cont'd)		
206.	Length to flange	Gage
207.	Length to shoulder	Gage
208.	Total length	Gage
209.	Length of slots	Gage
210.	Diameter of body, threaded end	Measure
211.	Diameter of flange	Measure
212.	Depth of effective thread	Measure
213.	Depth of threaded hole	Measure
214.	Length to angle of flange, min.	Measure
215.	Length of flat, min.	Measure
216.	Length to flange angle, max.	Measure
217.	Width across flat	Measure
218.	Marking missing, incorrect or illegible	Visual
219.	Radius or chamfer missing or incorrect	Visual
220.	Surface finish improper, except as otherwise classified	Visual

4.3.2.4 Guide, Drawing C8593282, covering a detail of Drawing F8593278.

CRITICAL

None defined

MAJOR

101.	Concentricity of boss cavity with cavity	Gage
102.	Concentricity of striker hole with cavity	Gage
103.	Concentricity of body with cavity	Gage
104.	True position of pinhole	Gage
105.	Diameter of cavity, max.	Gage
106.	Diameter of body, min.	Gage
107.	Diameter at datum location	Gage
108.	Diameter of pinhole, max.	Gage
109.	Height of boss	Gage
110.	Radius at root of boss cavity, missing or incorrect	Measure and Visual
111.	Metal defective	Visual
112.	Protective finish inadequate or defective	Visual
113.	Surface finish in cavity, improper	Visual
114.	Burr or sharp corners	Visual
115.	Foreign matter	Visual

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4.3.2.4 (Cont'd)

Defect		Method of Inspection
MINOR		
201.	Diameter of cavity, min.	Gage
202.	Diameter of body, max.	Gage
203.	Diameter of hole at front of cavity, min.	Gage
204.	Diameter of boss cavity	Gage
205.	Diameter of striker hole	Gage
206.	Depth of striker hole	Gage
207.	Total length	Gage
208.	Thickness through boss	Gage
209.	Location of pinhole	Gage
210.	Diameter of escapement hole at boss end	Measure
211.	Location of escapement hole at boss end	Measure
212.	Marking missing, incorrect or illegible	Visual
213.	Radius or chamfer missing or incorrect, except as otherwise classified	Visual
214.	Surface finish improper, except as otherwise classified	Visual

4.3.2.5 Pin, Firing, Drawing D8593864, covering a detail of Drawing F8593278.

CRITICAL

None defined

MAJOR

101.	Concentricity of groove with body	Gage
102.	Concentricity of striker with body	Gage
103.	Concentricity of striker shoulder with body	Gage
104.	True position of pinhole	Gage
105.	Diameter of groove	Gage
106.	Diameter of body, min.	Gage
107.	Diameter of striker, max.	Gage
108.	Diameter of pinhole, max.	Gage
109.	Depth of pinhole, max.	Gage
110.	Profile of striker	Gage
111.	Profile of groove	Measure
112.	Metal defective	Visual
113.	Protective finish inadequate or defective	Visual
114.	Surface finish of body and striker, improper	Visual
115.	Burr or sharp corner	Visual
116.	Foreign matter	Visual

MINOR

201.	Diameter of body, max.	Gage
202.	Diameter of striker, min.	Gage

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
203. Diameter of pinhole, min.	Gage
204. Diameter of striker shoulder	Gage
205. Depth of pinhole, min.	Gage
206. Length from body to tip of striker	Gage
207. Length of striker	Gage
208. Location of pinhole	Gage
209. Location of groove	Measure
210. Depth of slot in base	Measure
211. Length of flat, min.	Measure
212. Total length	Measure
213. Width of groove	Measure
214. Width of slot in base	Measure
215. Location of slot in base	Visual
216. Marking missing, incorrect or illegible	Visual
217. Radius missing or incorrect	Visual
218. Surface finish improper, except as otherwise classified	Visual

4.3.2.6 Pin, Shear, Drawing B8593284, covering a detail of Drawing F8593278.

CRITICAL	
None defined	
MAJOR	
101. Diameter, max.	Gage
102. Total length, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Surface finish improper	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
201. Total length, max.	Gage
202. Diameter of tapered end	Measure
203. Length of taper	Measure

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4.3.2.7 Piston, Drawing D8593285, covering a detail of Drawing F8593278.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of groove with shoulder	Gage
102.	Concentricity of shoulder with body	Gage
103.	Concentricity of pilot with body	Gage
104.	Concentricity of key locking groove with pilot	Gage
105.	Concentricity of escape groove with pilot	Gage
106.	Diameter of shoulder	Gage
107.	Diameter of pilot, min.	Gage
108.	Diameter of escape groove, max.	Gage
109.	Diameter of key locking groove	Gage
110.	Diameter of groove in large end, max.	Gage
111.	Dimaeter of body	Gage
112.	Profile of grooves	Measure
113.	Metal defective	Visual
114.	Protective finish inadequate or defective	Visual
115.	Surface finish on body and shoulder improper	Visual
116.	Burr or sharp corner	Visual
117.	Foreign matter	Visual
MINOR		
201.	Diameter of pilot, max.	Gage
202.	Diameter of escape groove, min.	Gage
203.	Diameter of groove in large end, min.	Gage
204.	Diameter of cavity	Gage
205.	Length of pilot	Gage
206.	Length to start of escape groove	Gage
207.	Length to shoulder	Gage
208.	Total length	Gage
209.	Width of flange, pilot end	Gage
210.	Depth of cavity	Measure
211.	Width of groove in large end	Measure
212.	Location of groove in large end	Measure
213.	Location of key escape groove	Measure
214.	Marking missing, incorrect or illegible	Visual
215.	Radius or chamfer, missing or incorrect	Visual
216.	Surface finish, improper, except as otherwise classified	Visual

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4.3.2.8 Collar, Drawing C8593286, covering a detail of Drawing F8593278.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Diameter outside, max.	Gage
102.	Diameter inside	Gage
103.	Width	Measure
104.	Thickness of wall, min.	Measure
105.	Metal defective	Visual
106.	Protective finish indequate or defective	Visual
107.	Surface finish, improper	Visual
108.	Burr or sharp corner	Visual
109.	Foreign matter	Visual
MINOR		
201.	Marking missing, incorrect or illegible	Visual
202.	Radii missing or incorrect	Visual

4.3.2.9 Spring, Piston Locking, Drawing C8593287, covering a detail of Drawing F8593278.

CRITICAL		
None defined		
MAJOR		
101.	Load at compressed length(s), incorrect	Gage
102.	Solid length, max.	Gage
103.	Ends incorrect	Visual
104.	Metal defective	Visual
105.	Protective finish inadequate or defective	Visual
106.	Burr or silver	Visual
107.	Foreign matter	Visual
MINOR		
201.	Diameter of wire	Measure
202.	Free length	Measure

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4.3.2.10 Key, Drawing C8593288, covering a detail of Drawing F8593278.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Radial profile of length	Gage
102. Radial profile of actual width	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Surface finish improper	Visual
106. Burr or sharp corners	Visual
107. Foreign matter	Visual
MINOR	
201. Total length	Gage
202. Radius missing or incorrect, except as otherwise classified	Visual

4.3.2.11 Metal Parts Assembly, Drawing F8593278, less all-components, except Firing Pin 8593864, Guide 8593282 and O-Ring MS 28775-012.

CRITICAL	
1. Protrusion of firing pin	Gage
MAJOR	
101. Any part missing	Visual
102. Improper assembly	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
NOne defined	

4.3.2.12 Metal Parts Assembly, Drawing F8593278.

CRITICAL	
None defined	

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4.3.2.12 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. End sleeve cannot be rotated 360 degrees clockwise	Manual
102. Any part missing	Visual
103. Improper assembly	Visual
104. Thread damaged	Visual
105. Marking incorrect, misleading or unidentifiable	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
201. Protective finish damaged or inadequate	Visual

4.3.2.13 Packing in Box, Fiberboard (Inner), Drawing C8796480.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Contents loose	Manual
202. Box broken or damaged	Visual
203. Number of units contained incorrect	Visual
204. Marking, missing, incorrect or illegible	Visual

4.3.2.14 Packing in Packing Box, Drawing F8796479.

CRITICAL

None defined

MAJOR

None defined

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4.3.2.14 (Cont'd)

Defect		Method of Inspection
MINOR		
201.	Contents loose	Manual
202.	Strapping missing, loose, broken or improperly assembled	Manual and Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking misleading or unidentifiable	Visual
207.	Number of units incorrect	Visual

4.3.2.14 Packing in Alternative Box Fiberboard (Outer), Drawing C9234882.

CRITICAL

None defined

MAJOR

None defined

MINOR

201.	Contents loose	Manual
202.	Box broken or damaged	Visual
203.	Number of units incorrect	Visual
204.	Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness.-Fifty units from each heat treated lot of components shall be tested for compliance with the values noted in the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.3.2 Hydrostatic pressure.-Each body and each head shall be subjected to the internal hydrostatic pressure. Failure of any unit to comply with the requirements of 3.5 shall be cause for rejection of the unit.

4.3.3.3 Shear pin strength.-Fifty pins from each lot shall be tested for shear strength. Failure of any shear pin to comply with the requirements of the drawing shall be cause for rejection of the lot.

4.3.3.4 Structural strength.-Each thruster shall be subjected to the structural strength test. Failure of any thruster to comply with the requirements of 3.6 shall be cause for rejection of the thruster.

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4.3.3.5 Protective finish.-The number of units for protective finish tests shall be in accordance with the applicable specifications. Failure of any sample to comply with the reqquirements of the applicable specifications shall be cause for rejection of the lot.

4.3.3.6 Residual magnetism.-Each assembly shall be subjected to a test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.9 shall be cause for rejection of the assembly.

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-8593278.

4.4 Test methods and procedures.

4.4.1 Hardness.-Prior to the application of protective finish, a hardness testing machine shall be used to determine the hardness values specified on the detail drawings.

4.4.2 Hydrostatic pressure.-Prior to the application of protective finish, the body and head shall be mounted in a device and subjected to an internal hydrostatic pressure in accordance with the requirements of 3.5. The testing device used shall internally seal the body and head, as applicable, at each end without applying any axial load other than that applied by the liquid pressure.

4.4.3 Shear pin strength.-After application of protective finish, the load shall be applied at the steady rate of 0.125 inch per minute. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.4.4 Structural strength.- The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.4.5 Protective finish.-Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing. Samples subjected to th protective finish test shall be scrapped.

4.4.6 Residual magnetism.-The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effect, by placing the thruster 6 inches from and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading, as shown in Figure 1. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90-degree rotation of the thruster.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be so specified by the procuring activity: Level A - in accordance with drawings C8796479 & C8796480; Level B - in accordance with drawings C9234882 and C8796480.

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6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Ordering data.-Procurement documents shall specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377- A581

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-60229A(MU) Thruster, Cartridge Actuated, M7 Metal Parts Assembly

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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